

4.3 Maximum & Minimum Values Worksheet

1. Find the absolute maximum and minimum values of the function $f(x) = x^3 - 3x^2 + 1$ on the interval

$$\left[-\frac{1}{2}, 4\right].$$

2. Find the absolute maximum and minimum values of $f(x) = x^3 - 12x + 1$ on the interval $[0, 5]$.

3. Find the absolute maximum and minimum values of the function $f(x) = (x^2 + x)^{2/3}$ on the interval $[-2, 3]$.

4. Find the absolute maximum and absolute minimum values of $f(x) = -2x\sqrt{x^2 + 9}$ on the interval $[-4, 1]$.

5. Find the absolute maximum and minimum values of $g(t) = t + \cot\left(\frac{t}{2}\right)$ on the interval $\left[\frac{\pi}{4}, \frac{7\pi}{4}\right]$.